

Total No. of Questions : 6]

SEAT No. :

PC-1960

[Total No. of Pages : 2

[6353] - 294

T.E. (Computer Engg.) (Honors)

VIRTUAL REALITY

(2019 Pattern) (Semester - I) (310701)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary

Q1) a) Explain in detail different geometric models? [9]

b) What is Orthographic projection and Perspective projection [8]

OR

Q2) a) What are axis angle representations of rotation? [9]

b) Explain different type of eye movements. [8]

Q3) a) Explain Rasterization in detail? [6]

b) Explain the structure of Human Eye? [6]

c) How to improve latency? [6]

OR

Q4) a) What is Visual Perception, Explain perception of Depth, Motion and Color? [6]

b) Explain the different ways to correct optical distortions? [6]

c) How to improve frame rates in Visual Perception? [6]

P.T.O.

- Q5)** a) Explain motion in Real and virtual world. [5]
b) Define velocities and acceleration. [4]
c) Explain Vestibular System in detail? [8]

OR

- Q6)** a) Explain Tracking position and orientation? [7]
b) Explain the difference between tracking 2D and 3D orientation? [10]
Q7) a) Explain Locomotion and Manipulation for interaction mechanism of VR. [6]
b) Explain the Auditory perception in detail? [6]
c) What is Auditory rendering? [6]

OR

- Q8)** a) Explain the physiology of human hearing. [6]
b) Explain the interaction with Motor programs and remapping of audio? [12]

